HDOC Day-10 - C Programming

Activities: Algorithm, Pseudocode, Flowchart, Test Cases, C Code, Compilation and Execution

Question 1: Convert Seconds into Days, Hours, Minutes and Seconds

Write a C program to input time in seconds and convert it into days, hours, minutes, and seconds format.

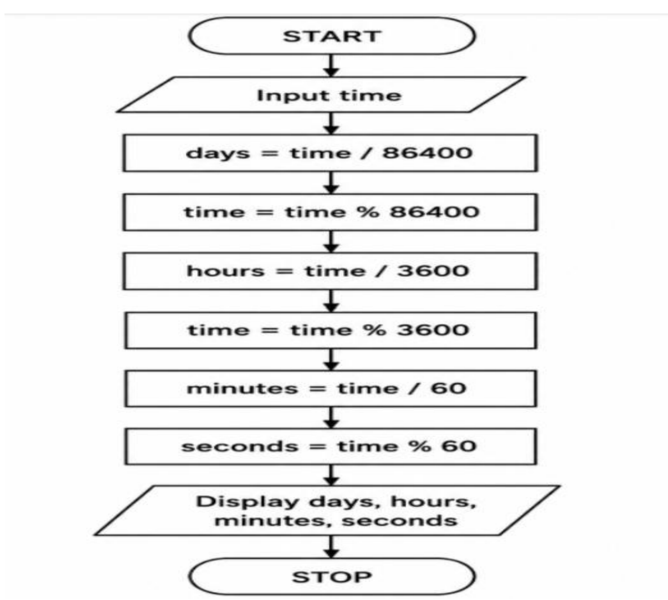
Algorithm

1. Start.
2. Input time in seconds.
3. Calculate days = time / 86400.
4. Keep the remaining seconds using time = time % 86400.
5. Calculate hours = time / 3600.
6. Keep the remaining seconds using time = time % 3600.
7. Calculate minutes = time / 60.
8. Calculate seconds = time % 60.
9. Display days, hours, minutes and seconds.
10. Stop.

Pseudocode

```
START
INPUT time
days = time / 86400
time = time % 86400
hours = time / 3600
time = time % 3600
minutes = time / 60
seconds = time % 60
DISPLAY days, hours, minutes, seconds
STOP
```

Flowchart



Test Case Table

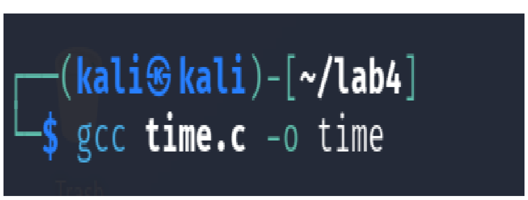
Test Case	Input (seconds)	Days	Hours	Minutes	Seconds
1	60	0	0	1	0
2	3661	0	1	1	1
3	86400	1	0	0	0
4	90061	1	1	1	1

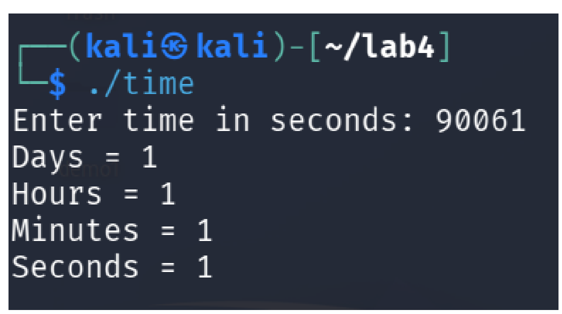
C Program

```
#include <stdio.h>

int main()
{
    int time, days, hours, minutes, seconds;

    printf("Enter time in seconds: ");
    scanf("%d", &time);

    days = time / 86400;
    time = time % 86400;

    hours = time / 3600;
    time = time % 3600;

    minutes = time / 60;
    seconds = time % 60;

    printf("Days = %d\n", days);
    printf("Hours = %d\n", hours);
    printf("Minutes = %d\n", minutes);
    printf("Seconds = %d\n", seconds);

    return 0;
}
```

Compilation and Execution

```
gcc time.c -o time
./time
```





Question 2: Check Leap Year

Write a C program to input a year and check whether it is a leap year or not.

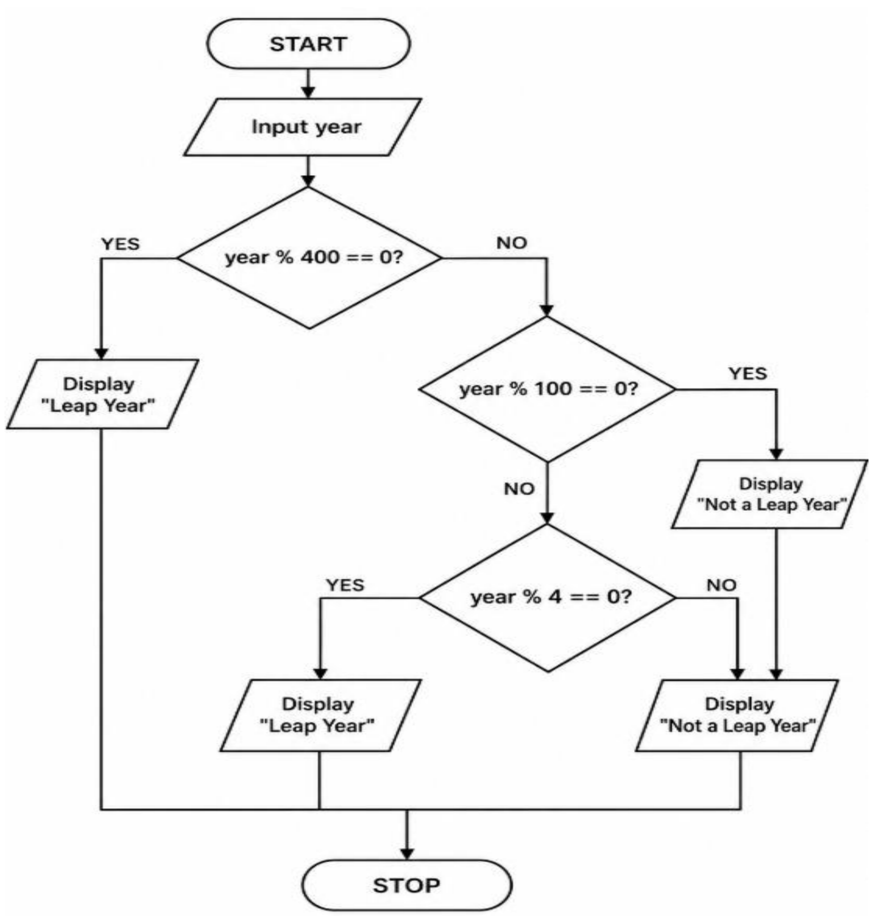
Algorithm

11. Start.
12. Input a year.
13. If the year is divisible by 400, display "Leap Year".
14. Else if the year is divisible by 100, display "Not a Leap Year".
15. Else if the year is divisible by 4, display "Leap Year".
16. Otherwise, display "Not a Leap Year".
17. Stop.

Pseudocode

```
START
INPUT year
IF year % 400 == 0
    DISPLAY "Leap Year"
ELSE IF year % 100 == 0
    DISPLAY "Not a Leap Year"
ELSE IF year % 4 == 0
    DISPLAY "Leap Year"
ELSE
    DISPLAY "Not a Leap Year"
STOP
```

Flowchart



Test Case Table

Test Case	Input Year	Expected Output
1	2024	Leap Year
2	2025	Not a Leap Year
3	1900	Not a Leap Year
4	2000	Leap Year

C Program

```
#include <stdio.h>

int main()
{
    int year;

    printf("Enter year: ");
    scanf("%d", &year);

    if (year % 400 == 0)
        printf("Leap Year");
    else if (year % 100 == 0)
        printf("Not a Leap Year");
    else if (year % 4 == 0)
        printf("Leap Year");
    else
        printf("Not a Leap Year");

    return 0;
}
```

Compilation and Execution

```
gcc leapyear.c -o leapyear
./leapyear
```